My health @ EU
eHealth Digital Service Infrastructure
A service provided by the European Union

My Health @ EU
eHealth Digital Service Infrastructure
Electronic cross-border health services in the EU

Service Location and Capability Lookup Profile

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ABSTRACT
This binding specifies the format and means for sharing information about eHealth DSI NCP provided services among eHealth DSI participating nations.

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V2.0.1	21/04/2017	Integrate the modifications linked to the CP-002	eHealth DSI provider	
V2.0.0	28/03/2017	Remove all references to epSOS and requirements	eHealth DSI provider	
V 1.0	20/01/2014	Publish	Fraunhofer FOKUS	ATNA

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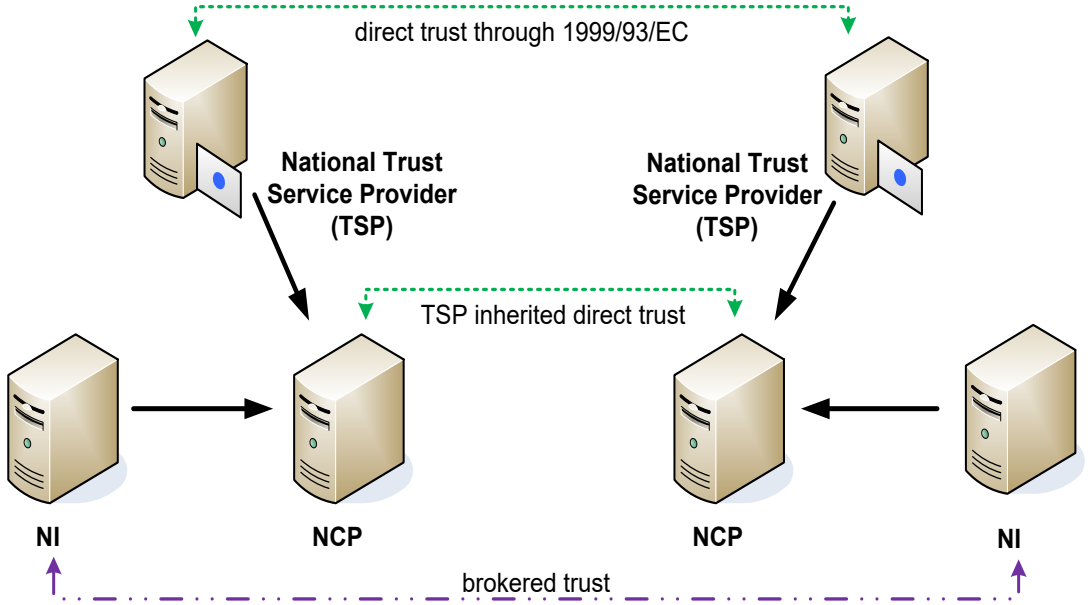
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1 Introduction

This binding specifies the format and means for sharing information about eHealth DSI NCP provided services among eHealth DSI participating nations.

1.1 eHealth DSI Service Metadata Publisher

eHealth DSI National Contact Point Gateways (NCPs) are operated by each eHealth DSI participating nation (PN). Each PN deploys the eHealth DSI functionalities for patient identification and document sharing on one or more secure nodes and by this offers respective services to other eHealth DSI PNs. The availability and location of eHealth DSI services as well as ancillary security artefact information for those NCP's are not statically provided for each PN but operate a sharing mechanism for dynamically publish and subscribe the current configuration.



In order to enable other PNs to access a PN's eHealth DSI services the service endpoints and digital certificates must be registered in a way that allows each PN to discover all other PN's service endpoints and verify the services' authenticity. For providing such a register, eHealth DSI makes use of "Service Metadata Publisher files" where each PN provides information about its managed services in a central service registry.

1.2 The ETSI TS 102 231 Standard

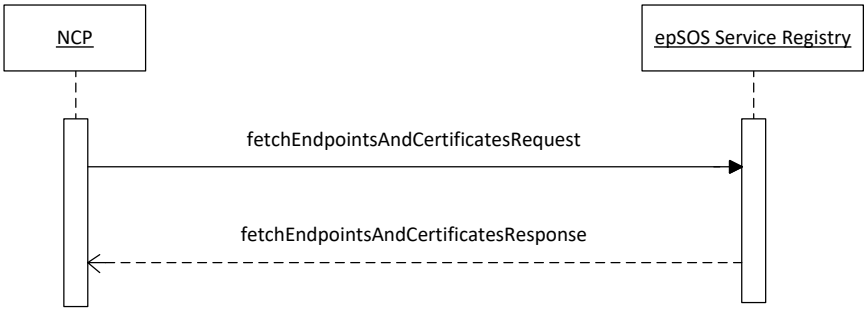
The purpose of the ETSI TS 102 231 standard "*is to provide a harmonized way in which assessment schemes having an oversight role with regards to trust services and their providers (Trust Service Providers - TSPs) can publish information about the services and TSPs which they currently oversee*" [ETSI TS 102 231].
For this the roles denoted in the quote above are mapped onto the eHealth DSI setting as follows:

ETSI TS 102 231	eHealth DSI
Assessment Scheme with on oversight role	Entity within an eHealth DSI PN that is responsible for eHealth DSI operation within that country (e.g. Ministry of Health)
Trust service provider	NCP operator

Services	NCP provided services (e.g. XCA responding gateway)
Information about the services	Endpoint address and digital certificate assigned to the service

1.3 The eHealth DSI SMP Registry

The eHealth DSI SMP registry is deployed as part of the eHealth DSI Central Services. For obtaining all eHealth DSI PN's EU Trust Lists defines a communication pattern that MUST be implemented by all NCPs.



Chapter 3 of this binding specifies the implementation of this communication pattern based on standard file transfer protocols.

1.4 Related Documents

The integration of the eHealth DSI Trusted Service Lists into the overall eHealth DSI Security Framework is defined in [Interoperability Specification] which is the normative foundation for all binding specifications. eHealth DSI participating nations' implementation of TLS data sets MUST make use of several security mechanisms and objects in order to fulfil the eHealth DSI security requirements. All cryptographic mechanisms and objects used for safeguarding eHealth DSI Trusted Service Lists MUST provide a level of technical security that MUST NOT be lower than the security level implemented through the normative eHealth DSI algorithm catalogue provided in [Cryptographic Algorithms].

1.5 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].
eHealth DSI requirements are managed within a central requirements management tool and serialized into [Requirements and Recommendations]. References to requirements always use the identifiers as used with the central requirements. Newly defined requirements are handed over to the central requirements management. In case of doubt or conflict, [Requirements and Recommendations] MUST be considered to be the normative phrasing of a requirement.

1.6 Terms and Definitions

The country which holds information about a patient, where the patient can be univocally identified and where his data may be accessed is called "country-A" (country of affiliation) [eHealth DSI Glossary]. The country of treatment where cross-border health care is provided when the patient is seeking care abroad is called "country-B" (country of care) [eHealth DSI Glossary].
A "National Contact Point (NCP)" is an entity in each particular country to act as a bidirectional technical, organizational and legal interface between the existing national functions and infrastructures [eHealth DSI Glossary]. In this document the term "NCP"

emphasizes the technical aspects of this interface and as such refers to a gateway which facilitates various aspects of cross-border data sharing (e.g. message forwarding, signature verification). From an architectural perspective NCPs denote the boundary between the eHealth DSI infrastructure and a country's existing national eHealth infrastructure.

1.7 Status of this Binding

The binding as defined in this document is a normative binding. All eHealth DSI participating nations MUST implement this binding within their NCP.
Additional or alternative bindings MUST NOT be defined and implemented for the eHealth DSI.

2 Download of eHealth DSI SMP from eHealth DSI Central Services

SMPs are published to the eHealth DSI central services by using a confidential and mutually authenticated protocol (such as sftp) to enforce confidentiality, integrity, and non-repudiation. Central Services move all the SMPs into a publicly available directory, where a synchronizer running in the NCP node can fetch. Once fetched, the SMP is authenticated and integrity checked. The synchronizer tool then auto-configures the NCP updating its configuration fields with the new values retrieved. Whenever a value changes, an audit trail is sent according with [Audit Trail Profiles].

2.1 FetchEndpointsAndCertificatesRequest Message (informative)

The currently exposed services of each eHealth DSI country may be configured for eHealth DSI as defined in the SMP-Editor User Manual.

2.2 Security Audit Considerations

The security Audit Considerations are defined in the section 2.3.5.3 of [Audit Trail Profiles].

3 eHealth DSI OASIS BDX-SMP/Location Binding

3.1 Capability Lookup through SML/SMP

Network addresses, web service endpoints and certificates of a country's eHealth DSI service providers and consumers are registered in a SMP SignedServiceMetadata structure, digitally signed by a trusted authority of the respective country (see [Security Services Specification] for details). SignedServiceMetadata of all countries are obtained by a NCP using the SMP+SML protocol set and it will be cached by the NCPs (caching strategies are not tackled by this specification).
SignedServiceMetadata are pointed by an overarching data structure, named ServiceMetadataCollection that holds all the services exposed by a NCP. Figure 2 and 3 shows the structure

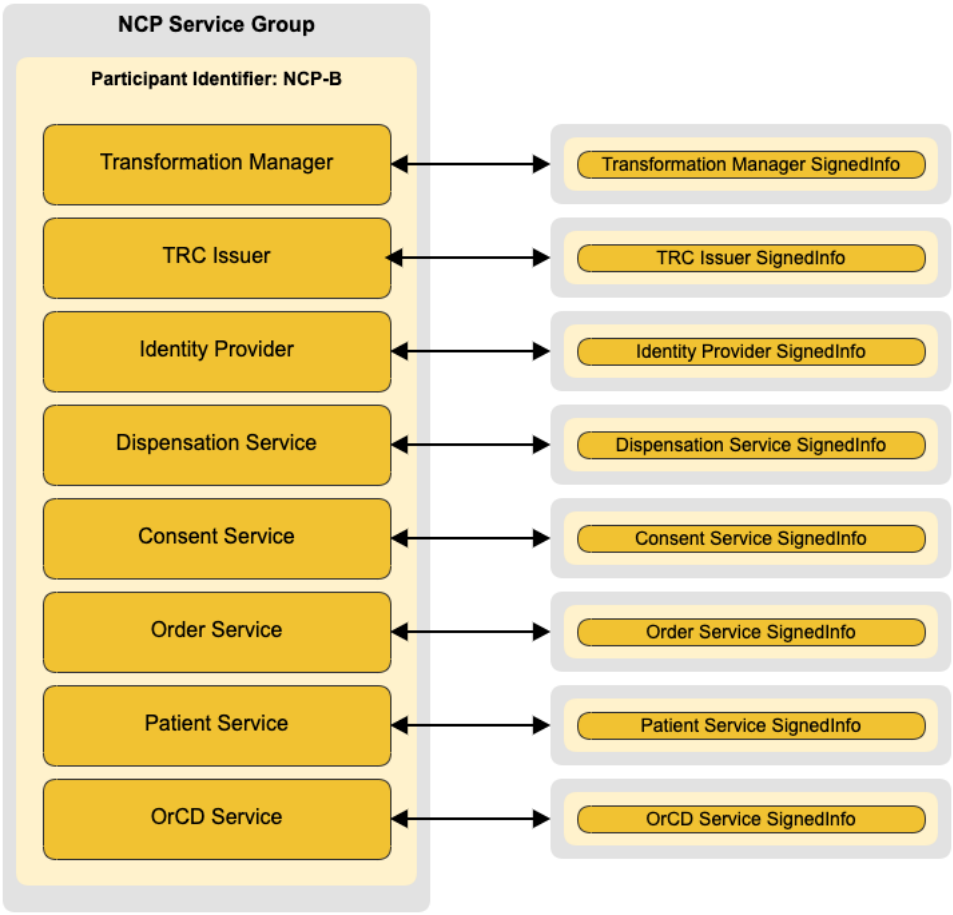


Figure 2: ServiceMetadataCollection data structure

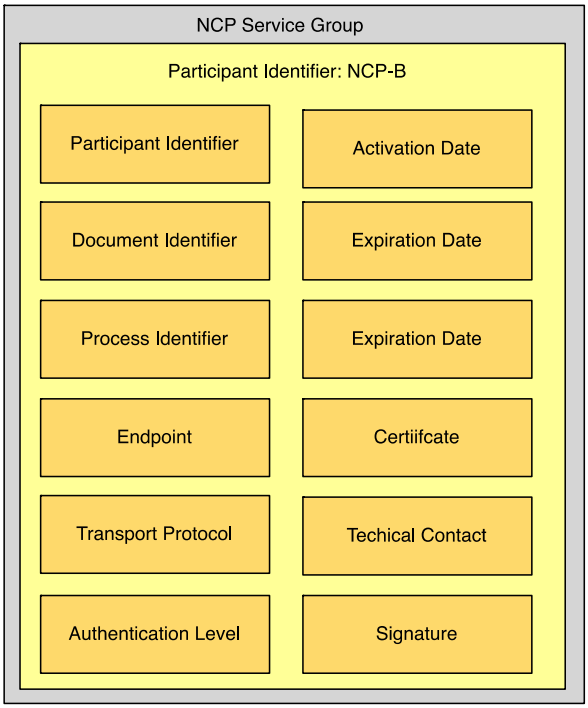


Figure 3: SignedServiceMetadata

The following sections define the application of the BDX-SMP for encoding the NCP configuration entries.

3.1.1 Normative Standards

This specification is based upon the following standards:

- http://docs.oasis-open.org/bdxr/BDX-Location/v1.0/BDX-Location-v1.0.html
- http://docs.oasis-open.org/bdxr/bdx-smp/v1.0/cs01/bdx-smp-v1.0-cs01.html
- [Audit Trail Profiles]

All the namespaces from the above documents apply.

3.1.2 Non-normative Standards

In addition to those in section 1.2, the following standards are considered to be informative, to support the implementation.

http://wiki.ds.unipi.gr/display/ESENS/PR+-+BDXL

3.1.3 ServiceGroup

The ServiceGroup is a file containing all the pointers for the SignedServiceMetadata, configuration repositories for the NCP's capabilities. Each entry on the service group points to a specific eHealth DSI service (e.g., patient service, order service).
The fields for the ServiceGroup MUST be used as follows for encoding the eHealth DSI services.

ServiceGroup Element	Opt	Usage Convention
ParticipantIdentifier	R	MUST be an unique URI of the NCP constructed as: "urn:ehealth:" + country code + "ncp<b|a>-idp".
ParticipantIdentifier/@Scheme	R	MUST be ehealth-participantid-qns
ServiceMetadataReferenceCollection	R	Holds the pointers to the service
ServiceMetadataReference	O	If present, it MUST contain the URL of the SignedServiceMetadata file associated for the service. It is usually set by the SMP implementation.

A sample mapping is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!-- This is the ServiceGroup file for the Portuguese NCPB -->
<ServiceGroup xmlns="http://docs.oasis-open.org/bdxr/ns/SMP/2016/05">
  <ParticipantIdentifier scheme="ehealth-participantid-qns">urn:ehealth:pt:ncpb-
idp</ParticipantIdentifier>
  <ServiceMetadataReferenceCollection>
    <!-- Request of Data - Query - XCA::CrossGatewayQuery - ITI-38 -->
    <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3ARequestOfData%3A%3AXCA%3A%3ACrossGatewayQuery%23%23I
TI-38" />
    <!-- Request of Data - Retrieve - XCA::CrossGatewayRetrieve - ITI-39 -->
    <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3ARequestOfData%3A%3AXCA%3A%3ACrossGatewayRetrieve%23%2
3ITI-39" />
    <!-- Request of Data - Fetch Request - XCF::CrossGatewayFetchRequest - ITI-63 -->
    <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3ARequestOfData%3A%3AXCF%3A%3ACrossGatewayFetchRequest%
23%23ITI-63" />
    <!-- Provisioning of Data - Provide - XDR::ProvideandRegisterDocumentSet-b - ITI-41 -->
    <ServiceMetadataReference href="http://smp-digit-
```

```
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3AProvisioningOfData%3AProvide%3A%3AXDR%3A%3AProvideandRe
gisterDocumentSet-b%23%23ITI-41" />
   <!-- Provisioning of Data - BPPC-RegisterUpdate - XDR::BPPCProvideandRegisterDocumentSet-b -
ITI-41 -->
   <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3AProvisioningOfData%3ABPPC-
RegisterUpdate%3A%3AXDR%3A%3AProvideandRegisterDocumentSet-b%23%23ITI-41" />
   <!-- Patient Identification and Authentication - XCPD - XCPD::CrossGatewayPatientDiscovery - ITI-55
-->
   <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3APatientIdentificationAndAuthentication%3A%3AXCPD%3A%3ACross
GatewayPatientDiscovery%23%23ITI-55" />
   <!-- Country-B Identity Provider - Issuance of HP Identity Assertion - XUA::ProvideX-UserAssertion -
ITI-40 -->
   <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3A+CountryBIdentityProvider%3A%3AXUA%3A%3AProvideX-
UserAssertion%23%23ITI-40" />
   <!-- ISM - International Search Mask - ehealth-107 -->
   <ServiceMetadataReference href="http://smp-digit-
mock.publisher.ehealth.acc.edelivery.tech.ec.europa.eu/ehealth-participantid-
qns%3A%3Aurn%3Aehealth%3Apt%3Ancpb-idp/services/ehealth-resid-
qns%3A%3Aurn%3Aehealth%3AISM%3A%3AInternationalSearchMask%23%23ehealth-107" />
  </ServiceMetadataReferenceCollection>
</ServiceGroup>
```

NCP-A signature certificate is the urn:ehealth:CountryBIdentityProvider::identityProvider::HPAuthentication##epsos-91/Certificate.

The self-assessment of client certificate is optional and it can be achieved the same way, by changing the participant identifier of the service group to either NCP-B or A. Trust bootstrap is out of scope of this specification and achieved by, e.g., TL, see https://www.enisa.europa.eu/topics/trust-services/guidelines. Remember that the CAs are Trusted Service Provider.

The SMP server is not considered as a qualified electronic registered delivery service. For this reason, the SMP server itself can't be used to perform trust bootstrap. Moreover the scheme operator can't vouch for a TSP in a country, this is made by another organizational body (usually some group in the MoI). Thus the CAs are listed in the Trusted List, and maintained and kept up-to-date by another body. If there is a change in the state of qualification, a separate process will advise the NCP operator (see e.g., the ENISA publication above). E.g., during installation time, the operator will ask for the TL, validate the TL, and install it in the truststore.

In this example the Service Group is created for the Portuguese NCP. Each MetadataReference points to the document containing the SignedServiceMetadata.

3.1.4 SignedInformation

The SignedInformation file contains the information related to the specific eHealth resource (e.g., eHealth DSI service). The following encoding MUST apply.

SignedServiceMetadata	Opt	Usage Convention
ParticipantIdentifier	R	MUST be an unique URI of the NCP constructed as: "urn:ehealth:" + country code + "ncpb-idp".
ParticipantIdentifier/@Scheme	R	MUST be ehealth-participantid-qns

DocumentIdentifier	R	It MUST contain a string following the format specified in section 2.4.6.1 of OASIS BDX-SMP specification and including the specific audit Trail eHealth DSI EventIDs / ITIs" DocumentIdentifier SHALL be composed as "urn:ehealth:"+interaction pattern+"::"+event name+"##"+EventID / Transaction and ProcessIdentifier as "urn:" + event name, according with table 2.3.5.7 of [Audit trail Profiles]
DocumentIdentifier/@Scheme	R	It MUST be ehealth-resid-qns
ProcessList	R	It holds the processes that provide services for the specific service
Extension	R	It MUST contain the signature of the national authority who created the file (e.g., the scheme operator).

Each process is encoded as follows.

Process	**Opt**	**Usage Convention**
ProcessIdentifier	R	MUST contain the URI of the specific service, encoded as EventName according with Section 2.3.5.7 of [Audit Trail profiles]
ProcessIdentifier/@Scheme	R	MUST be ehealth-procid-qns
ServiceEndpointList	R	MUST contain the service information for each service endpoint served by this service. E.g., if service is Patient Service, the operation is List().
Endpoint/@transportProfile	R	MUST contain the selected profile for this service. Available profiles are: • urn:ihe:iti:2013:xcpd • urn:ihe:iti:2013:xds • urn:ihe:iti:2013:xca • urn:ihe:iti:2013:xcf Applications MAY add additional protocols (e.g., for eID)
EndpointURI	R	MUST contain the WSE of the specific service
RequireBusinessLevelSignature	R	MUST be set to false (as it is not used by eHealth)
MinimumAuthenticationLevel	O	MAY contain the minimum Level of Authentication (LoA, AAL) required to access to the service.
ServiceActivationDate	R	MUST contain the Date when the service has started
ServiceExpirationDate	O	MAY contain the Date when the service will be stopped
Certificate	R	MUST contain the public certificate used to run the service
ServiceDescription	O	MAY contain the English description of the service
TechnicalContactURL	O	MAY contain the information related to the technical contact
TechnicalInformationURL	O	MAY contain the URL pointer to the remote service technical description
Extension	O	MAY contain additional service-specific extension (processcontent = LAX)

A sample SignedServiceMetadata file for the Patient Identification transaction is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<SignedServiceMetadata xmlns="http://docs.oasis-open.org/bdxr/ns/SMP/2016/05">
 <ServiceMetadata xmlns:ns0="http://www.w3.org/2000/09/xmldsig#">
  <ServiceInformation>
   <ParticipantIdentifier scheme="ehealth-participantid-qns"
    >urn:ehealth:pt:ncpa-idp</ParticipantIdentifier>
   <DocumentIdentifier scheme="ehealth-resid-qns"
    >urn:ehealth:PatientIdentificationAndAuthentication::XCPD::CrossGatewayPatientDiscovery##ITI-
55</DocumentIdentifier>
```

```
    <ProcessList>
     <Process>
      <ProcessIdentifier scheme="ehealth-procid-qns"
       >urn:XCPD::CrossGatewayPatientDiscovery</ProcessIdentifier>
      <ServiceEndpointList>
       <Endpoint transportProfile="urn:ihe:iti:2013:xcpd">
        <EndpointURI>https://openncp.pt:8443/services/xcpd</EndpointURI>
        <RequireBusinessLevelSignature>false</RequireBusinessLevelSignature>
        <MinimumAuthenticationLevel>urn:epSOS:loa:1</MinimumAuthenticationLevel>
        <ServiceActivationDate>2017-02-01T00:05:00Z</ServiceActivationDate>
        <ServiceExpirationDate>2017-02-28T23:55:00Z</ServiceExpirationDate>
<Certificate>MIIFNjCCAx6gAwIBAgIJAOO4WCYoxzPWMA0GCSqGSIb3DQEBDQUAMIGMMQswCQYDVQQGEwJQVDEOMAwGA1
UECAwFUG9ydG8xDjAMBgNVBAcMBVBvcnRvMQ0wCwYDVQQKDARTUE1TMQwwCgYDVQQLDANJT1AxEzARBgNVBAMMCmpv
YW8uY3VuaGExKzApBgkqhkiG9w0BCQEWHGpvYW8uY3VuaGFAc3Btcy5taW4tc2F1ZGUucHQwHhcNMTYwOTEzMTU0NDAyWh
cNMTcwOTEzMTU0NDAyWjBaMQswCQYDVQQGEwJQVDENMAsGA1UEChMEU1BNUzENMAsGA1UEKhMESm9hbzEOMAwGA1
UEBBMFQ3VuaGExHTAbBgNVBAMTFHFhZXBzb3MubWluLXNhdWRlLnB0MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQ
EA0JSJWVCyaLeq5AcI5j8cd82pI8hRTPIbfPLmQijWkMxHjOieKVsEet1ATOFU9/QNcluILD+1+2gBAngsP6dqv1ctV9nhxrTzTU5ms9jt
lu+ex5KRFtK7Z4h9cuKKRVdvZouFZCxgxhESKUY8IRzUSkfOAe5Df3qQgvmxJeFreRk5n+5YN+0d4aBf+Vt9E9WzYrqQESojVZ5LkE1d
MOFIEwnc7M7iRpPkZD7Cb41/dZ0PuX4DjpQkaAASzlcNzuF73lr+1tPHb6OiyunUQYY8IHj06NechnlrrqexbBYqhp1UhL4IV2/xUx8u
yq2BFiZhiebCxUDtK+YKyTCWD6l3gQIDAQABo4HLMIHIMCcGA1UdEQQgMB6BHGpvYW8uY3VuaGFAc3Btcy5taW4tc2F1ZGUucH
QwDgYDVR0PAQH/BAQDAgbAMB8GA1UdIwQYMBaAFHWz6cJ961Cgvw53QD58LN7XzxZcMAwGA1UdEwEB/wQCMAAwHQYDV
R0OBBYEFMQilafrh9brFb8jXUh+9STmgMUpMD8GA1UdHwQ4MDYwNKAyoDCGLmh0dHA6Ly95b3VyLmNlcnQuYXV0aG9yaXR5
L0NSTC9vcGVubmNwLWNybC5wZW0wDQYJKoZIhvcNAQENBQADggIBACycmMwqEqL2/PMKnc1My0WablbzUy6wjJY49vvmMR
dX6k/4USBR/Y8oZEmvmXxE6AYeZRlwAqRMPpH4zXk0zO74RbuVchHM5L173F2P91qFgLgyFEDDOjII17zQeEEYYL78eMLswnqk4X
LYrchOG/Ql+cXhMJafELEU4VLiJnFWcnb1D+spMXobO5ramOdZwVMvbQJUqRXqPvoJ4ZYhktoW+p2nPQahbDuv55FympVEzFJBP
MUvGIanLMauTKcMMZzEIUch3VyBqXyDzj2SqajKvAUVoU38lNauDwe9PujcmvaW0J/6RJ7SbkU20j1Y/urSn6QBMJfsHW2LwX5cto
+ulacXUHjcWZhimaJJ86atjAfDwWE6y0f/ENSzi8eZ693hywmRgU6te7H4lOzUzd3exMbiFNxVTfmMHImou8ZBHUb7wiVUNShc/Q9
F1gppTpUwd8gV74b3ZeyUCZ6o7EDX/OSdZNbPxRlo+cNrQ0Gp8JoWEJ3R6kJzMhAMs/DYwUAfKyfocekhaUEIxFBu4JP5rR03a60r
EjI3v5mHWutbn86g2b2TrGssl/ZsLJxyIBM1vDSJ75gGYIwzje/QmKaSGcnr1RJo5Z6DG25Tf0kbWUWJxMyGj8dSnsfHkoiTU4mfcoZc
amfcnXGeHm0vLvTS3j3G31fo1uZMIJ53V5r5</Certificate>
        <ServiceDescription>This is the Patient Identification and
         Authentication configuration for the PT NCP</ServiceDescription>
        <TechnicalContactUrl>xyz@email.com</TechnicalContactUrl>
        <TechnicalInformationUrl>http://smps.pt/ncp/information</TechnicalInformationUrl>
       </Endpoint>
      </ServiceEndpointList>
     </Process>
    </ProcessList>
    <Extension>
     <Signature xmlns="http://www.w3.org/2000/09/xmldsig#">
      <SignedInfo>
       <CanonicalizationMethod
        Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"/>
       <SignatureMethod Algorithm="http://www.w3.org/2000/09/xmldsig#rsa-sha1"/>
       <Reference URI="">
        <Transforms>
         <Transform
          Algorithm="http://www.w3.org/2000/09/xmldsig#enveloped-signature"
         />
        </Transforms>
        <DigestMethod Algorithm="http://www.w3.org/2001/04/xmlenc#sha256"/>
        <DigestValue>H7vGNy+okvOsdZrVgvD8pfM4XrzsHjV1wi8tELbtoOQ=</DigestValue>
       </Reference>
      </SignedInfo>
      <SignatureValue>oHx787zVXJX5Qar7Fhipf+d0oOcZF5QSNUCMZrU+n/qdmBLJZDpmCajBVriwPMhu+sa3kP/wDM6c
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       slWn3W1yMT4mWed/gW8=</SignatureValue>
      <KeyInfo>
       <X509Data>
        <X509SubjectName>CN=Sample National
         Infrastructure,OU=Sante,C=PT</X509SubjectName>

<X509Certificate>MIICAzCCAWygAwIBAgIEWCRzHjANBgkqhkiG9w0BAQsFADBGMQswCQYDVQQGEwJQVDEOMAwGA1UE
```

```
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        MCUGA1UEAwweU2FtcGxlIE5hdGlvbmFsIEluZnJhc3RydWN0dXJlMIGfMA0GCSqGSIb3DQEBAQUA
        A4GNADCBiQKBgQCywt50WXEWIiWytRGcMqzeMM/EyxruNthPdiUEUTbs9un7lzGGjpfFMTgd83wJ
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        L4XH1GcdCg6oPKPUc+ovbgN7/iR265Elp4qHfpVteBijBTyZReH4oAK9hRhK1gLwtjI7vpjVaPXv
        vkV1fbrz</X509Certificate>
      </X509Data>
    </KeyInfo>
   </Signature>
  </Extension>
 </ServiceInformation>
</ServiceMetadata>
<Signature xmlns="http://www.w3.org/2000/09/xmldsig#">
 <SignedInfo>
  <CanonicalizationMethod Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"/>
  <SignatureMethod Algorithm="http://www.w3.org/2000/09/xmldsig#rsa-sha1"/>
  <Reference URI="">
   <Transforms>
    <Transform Algorithm="http://www.w3.org/2000/09/xmldsig#enveloped-signature"/>
   </Transforms>
   <DigestMethod Algorithm="http://www.w3.org/2001/04/xmlenc#sha256"/>
   <DigestValue>Fk7J0BqaRUuNQ/zupaek1sWMvMzbciY7Cf8SKiKwUjQ=</DigestValue>
  </Reference>
 </SignedInfo>
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  GQ1iJufsdLQFKMh0acBHqaMbeD8y6gJqD5cFsvMmPEPDVcFd6yh2zqCTalaxfLJp5gUSE/SGetih
  N/tcmpg2Rp5YinLo5l4=</SignatureValue>
 <KeyInfo>
  <X509Data>
   <X509SubjectName>CN=SMP Mock Services,OU=DIGIT,O=European
    Commision,C=BE</X509SubjectName>
   <X509Certificate>MIICIzCCAYygAwIBAgIEWCRzfjANBgkqhkiG9w0BAQsFADBWMQswCQYDVQQGEwJCRTEbMBkGA1UE
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    MBkGA1UECgwSRXVyb3BlYW4gQ29tbWlzaW9uMQ4wDAYDVQQLDAVESUdJVDEaMBgGA1UEAwwRU01Q
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    K4XRn2+doLjN2dHremGyeKBgLb0Ulp9E9I5P8kxuIs7TjroxZofK9ixhfBv5rJhLcHy8XdrUYqAS
    awc3c5bM9fNxRWCMkNYNoSYVxPBdlS4zEeLNNzRY+wjrMNYIJR4=</X509Certificate>
  </X509Data>
 </KeyInfo>
</Signature>
</SignedServiceMetadata>
```

Every ServiceMetadata MUST be electronically signed by its scheme operator in compliance with the electronic signature and Trusted List requirements as laid out in 910/2014/EC. An electronic seal may be used in order to enable legal persons to apply such safeguards onto the data, however, not before specifically acknowledging the different legal effect of an electronic seal in comparison to an electronic signature. The electronic signature or seal MUST be applied to *ServiceMetadata/ServiceInformation/Extension/ds:Signature* and its security level MUST be coherently correspond with the protection demands of to-be safeguarded information. The SignedServiceMetadata MUST be signed by the SMP server adhering to the aforementioned requirements. This electronic signature or seal MUST be applied to *SignedServiceMetadata/ds:Signature*. Electronic signatures or seals MUST spawn the entire payload of the document they are electronically signing.

3.1.5 NCP Provider Identification

The ServiceGroup element MUST contain a single ServiceMetadataReferenceCollection element for each face of the NCP (NCP-A and/or NCP-B) that is operated by the respective member state.
The service list for each face of the NCP contains entries for the gateways and services of this NCP. The following table shows, which service entries MUST be filled (R), which MAY be filled (O) and which service entries MUST NOT be used (X) for service providers (NCP-A) and service consumers (NCP-B).

Gateway / Service	Opt. NCP-A	Opt. NCP-B	Reference
NCP	R	R	
Patient Identification Service	R	X	
Patient Service	O	X	
Order Service	O	X	
Dispensation Service	O	X	
Consent Service	O	X	
HCP Identity Provider	X	O	
HCP Signature CA	O	O	

3.1.6 Use of Dedicated eHealth DSI Identity Providers

Country-B implementations MAY use dedicated Identity Providers within NCP-B for issuing HP Identity Assertions. In this scenario the HP Identity Assertion MUST be signed by the Identity Provider. Identity Provider status information can be used for distributing the Identity Provider certificate.

Process	Opt	Usage Convention
ProcessIdentifier	R	
ProcessIdentifier/@Scheme	R	MUST be ehealth-procid-qns
ServiceEndpointList	R	
Endpoint/@transportProfile	R	The transport profile used to validate an assertion. It MAY be one of • WS-Trust 1.3 - http://docs.oasis-open.org/ws-sx/ws-trust/200512 • SAML Protocol v2 - urn:oasis:names:tc:SAML:2.0:protocol • OAuth 2.0 - rfc6749
EndpointURI	R	Fully qualified domain names and/or IP-addresses of the dedicated IDP
RequireBusinessLevelSignature	X	
ServiceActivationDate	R	MUST contain the Date when the service has started
ServiceExpirationDate	R	MUST contain the Date when the service will be stopped
Certificate	R	Certificate that this IDP is using to sign
ServiceDescription	O	MAY contain the english description of the service
TechnicalContactURL	O	MAY contain the information related to the technical contact
TechnicalInformationURL	O	MAY contain the URL pointer to the remote service technical description
Extension	O	MAY contain additional service-specific extension (processcontent = LAX) For instance, the list of trusted remote IdPs.

3.1.7 International Search Mask

eHealth portals use an adaptive International Search Mask configuration to switch the presentation of the portal for the specific patient's country. Since the International Search Mask is not defined in [Audit Trail Profiles], the following document identifier shall apply:

- ehealth-107

Process	Opt	Usage Convention
ProcessIdentifier	R	MUST contain "urn:ehealth:ncp:<country>:ism"
ProcessIdentifier/@Scheme	R	MUST be ehealth-procid-qns

ServiceEndpointList			R	
	Endpoint/@transportProfile		R	MUST be "urn:ehealth:transport:none"
		EndpointURI	X	
		RequireBusinessLevelSignature	X	
		ServiceActivationDate	R	MUST contains the Date when the mask has been issued
		ServiceExpirationDate	O	MUST contains the Date when the service will be stopped
		Certificate	X	
		ServiceDescription	O	MAY contain the english description of the search mask
		TechnicalContactURL	O	MAY contain the information related to the technical contact
		TechnicalInformationURL	O	MAY contain the URL pointer to the remote mask technical description
	Extension		R	MUST contain the international search mask XML as direct children

3.1.8 International Search Mask Extension

ISM extension SHALL be defined by each eHNCP-A to allow identification of patients and documents according each National policy.

Any attribute has the possibility to configure the scope of their use (0=all, 1=Patient Summary, 2 = Prescription/Dispense, default value is 0).

For translation purpose, each attribute contains a label key which SHALL be filled in by value from the value set "Standard Data Name" from [eHDSI Requirements Catalogue]

- Patient Search:
  - Country
  - LivingSubjectId (National ID, Regional, Passport etc.)
  - FamilyName (patient.data.surname)
  - FirstName (patient.data.givenname)
  - Gender (patient.data.sex)
  - BirthDate (patient.data.birth.date)
  - AddressStreet (patient.data.street.address)
  - AddressCity (patient.data.city)
  - AddressPostalCode (patient.data.code)
  - AddressCountry (patient.data.country)

- Next of Kin (optional):
  - Country
  - LivingSubjectId (National ID, Regional, Passport etc.)
  - FamilyName (patient.data.surname)
  - FirstName (patient.data.givenname)
  - Gender (patient.data.sex)
  - BirthDate (patient.data.birth.date)
  - AddressStreet (patient.data.street.address)
  - AddressCity (patient.data.city)
  - AddressPostalCode (patient.data.code)
  - AddressCountry (patient.data.country)

- Document Search (optional):
  - prescriptionId: (ePrescriptionId)
  - pinCode (dispensationPINCode)

- Media:
  - Picture base 64 encoded (media)

Refer the international seach mask schema definitions in OpenNCP (code.europa)

4 References

4.1 Normative References

[ETSI TS 102 231] ETSI: Electronic Signatures and Infrastructures (ESI); Provision of harmonized Trust-service status information. Version 3.1.2, December 2009. http://www.etsi.org/deliver/etsi_ts/102200_102299/102231/03.01.02_60/ts_102231v030102p.pdf

[RFC 2119] Bradner, S.: Key words for use in RFCs to Indicate Requirement Levels. Harvard University, Boston, Massachusetts, 1997.